

CHAPTER 18: NEURAL CONTROL AND COORDINATION

Comprehensive Academic Notes & Master Study Guide • 100% NCERT Content Preservation • Landscape Format

Exam Target: NEET / CBSE / Boards

Includes All Definitions, Formulas & Diagrams

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CHAPTER MIND MAP — NEURAL CONTROL AND COORDINATION

NEURAL CONTROL & COORDINATION (HOMEOSTASIS)

1. NEURAL ORGANISATION

- **Invertebrates:**
 - Hydra: Simple Neural Network

2. HUMAN NEURAL SYSTEM

- **CNS:**
 - Brain & Spinal Cord (Processing)

3. NEURON & PHYSIOLOGY

- **Structure:**
 - Cell body, Dendrites, Axon

4. CENTRAL NEURAL SYSTEM (CNS)

- **Meninges:**
 - Dura, Arachnoid, Pia mater

- Insects: Brain + Ganglia + Tissues

• **Vertebrates:**

- Highly Developed Neural System

• **PNS:**

- Afferent (Sensory) & Efferent (Motor)

- Somatic (Skeletal) & Autonomic (Visceral)

- Sympathetic & Parasympathetic

• **Conduction:**

- Resting Potential (-70mV, Na⁺/K⁺ Pump)

- Action Potential (+30mV, Na⁺ Influx)

• **Synapse:**

- Electrical vs Chemical (Neurotransmitters)

• **Forebrain:**

- Cerebrum, Thalamus, Hypothalamus

• **Midbrain:**

- Corpora Quadrigemina, Aqueduct

• **Hindbrain:**

- Pons, Cerebellum, Medulla

18.1 COORDINATION & NEURAL SYSTEM

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DEFINITION: COORDINATION

Coordination is the physiological process through which two or more organs interact and complement the functions of one another to maintain **Homeostasis** (steady internal state).

Exercise Example (Integrated Physiology):

- **Increased Physical Activity** → Higher muscular energy demand → Increased O_2 requirement.
- **Systemic Response:** Increased rate of respiration + Increased heart beat + Increased blood flow via blood vessels.
- **Post-Exercise Recovery:** Nerves, lungs, heart, and kidneys gradually return to normal resting states in a synchronized fashion.

Neural System

Provides an organized network of point-to-point connections for **rapid / quick coordination**.

Endocrine System

Provides chemical integration through **hormones** (slower, long-lasting regulation).

Comparative Neural Organization across Animal Kingdom:

- **Lower Invertebrates (e.g., Hydra):** Composed of a simple network of neurons.
- **Insects:** Better organized; possess a **brain** along with a number of **ganglia** and neural tissues.
- **Vertebrates:** Possess a highly developed and complex neural system.

18.2 HUMAN NEURAL SYSTEM STRUCTURE

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HUMAN NEURAL SYSTEM

1. Central Neural System (CNS)

Includes **Brain + Spinal Cord**. Primary site of information processing and control.

2. Peripheral Neural System (PNS)

All nerves of body associated with CNS.

PNS Fibre Type	Direction of Impulse Flow	Functional Target
Afferent Fibres (Sensory)	Tissues / Organs → CNS	Transmits sensory inputs to processing center
Efferent Fibres (Motor)	CNS → Peripheral Tissues / Organs	Transmits regulatory motor commands to effectors

Somatic Neural System (SNS)

Relays impulses from CNS to **skeletal muscles** (Voluntary movement control).

Autonomic Neural System (ANS)

Transmits impulses from CNS to **involuntary organs & smooth muscles**.

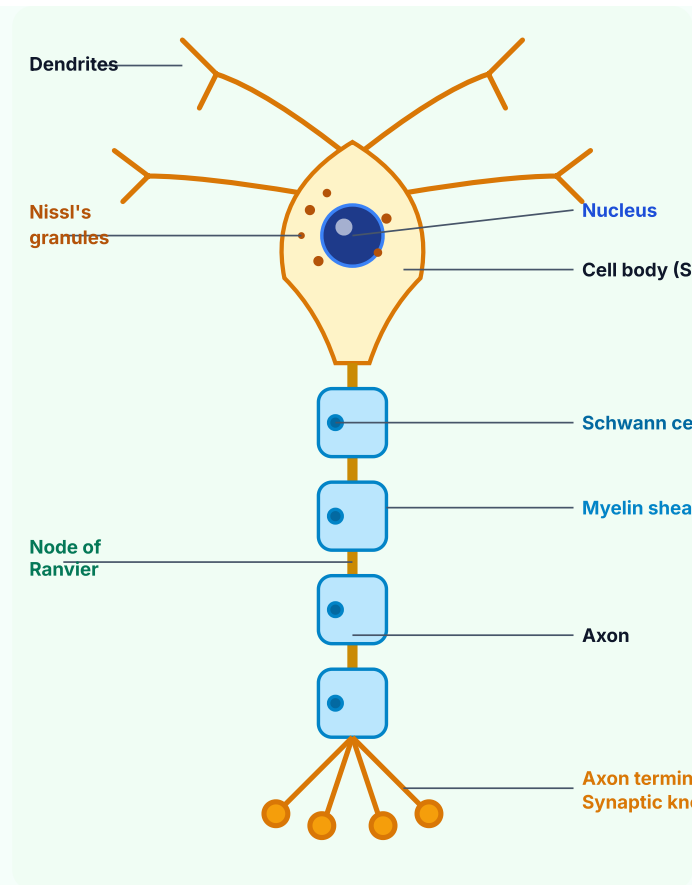
Subdivisions: (i) Sympathetic & (ii) Parasympathetic

Visceral Nervous System: Part of PNS comprising nerves, fibres, ganglia, and plexuses by which impulses travel between CNS and the viscera (internal organs).

18.3 NEURON — STRUCTURAL AND FUNCTIONAL UNIT

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FIGURE 18.1 — DETAILED STRUCTURE OF A NEURON



1. Cell Body (Soma)

Contains cytoplasm with typical cell organelles and granular bodies called **Nissl's granules**.

2. Dendrites

Short, repeatedly branched fibres projecting out of cell body. Contain Nissl's granules. Transmit impulses **towards cell body** (centripetal).

3. Axon

Long single fibre with branched distal end. Terminates in bulb-like **Synaptic Knobs** containing **neurotransmitters**. Transmits impulses **away from cell body** (centrifugal).

Structural Types of Neurons (Based on number of Axons & Dendrites)

Neuron Type	Structural Composition	Specific NCERT Anatomical Location
Multipolar	One Axon + Two or more Dendrites	Cerebral Cortex
Bipolar	One Axon + One Dendrite	Retina of Eye
Unipolar	Cell body with One Axon only	Usually found in Embryonic Stage

⚡ Myelinated Nerve Fibres

- Enveloped with **Schwann cells** that form a **myelin sheath** around the axon.
- Gaps between adjacent sheaths = **Nodes of Ranvier**.
- **Found in:** Spinal and Cranial Nerves.

🕯 Unmyelinated Nerve Fibres

- Enclosed by Schwann cells that **DO NOT form a myelin sheath** around the axon.
- No Nodes of Ranvier.
- **Found in:** Autonomous and Somatic neural systems.

18.3.1 GENERATION AND CONDUCTION OF NERVE IMPULSE

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MATHEMATICAL & ION GRADIENT MECHANICS OF AXONAL MEMBRANE

Physiological Constants

1. Active Transport Pump Ratio:

Sodium-Potassium Pump ($3\text{Na}^+ / 2\text{K}^+ \text{ ATPase}$)

Transports: **3 Na^+ OUTWARDS** for **2 K^+ INWARDS** (Uses 1 ATP)

2. Resting Potential Equation:

Resting Membrane Potential $\approx -70\text{ mV}$

Inner surface: **Negative (-)** | Outer surface: **Positive (+)**

3. Action Potential Equation:

Action Potential (Nerve Impulse) $\approx +30\text{ mV}$

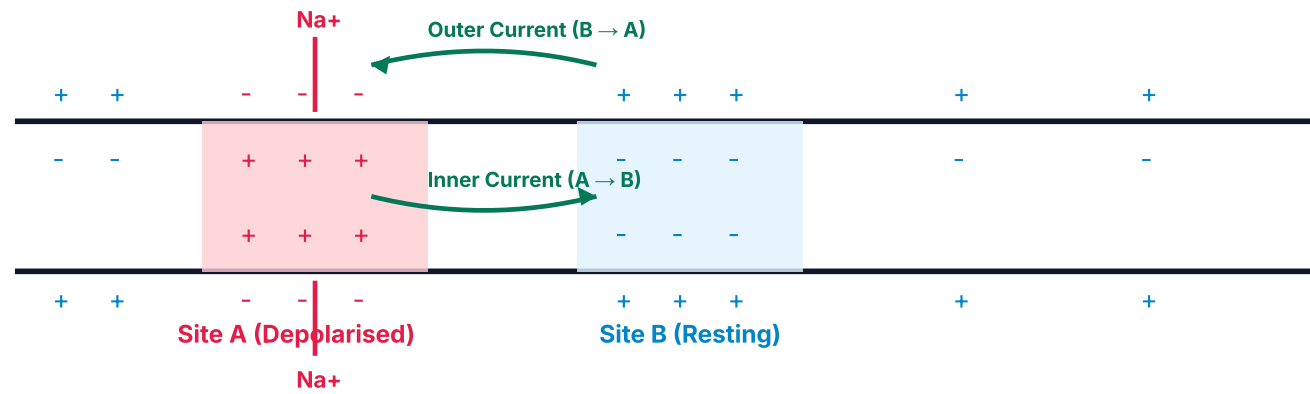
STAGE 1: Resting State (Polarised Membrane)

- 1 **Selective Permeability:** Axonal membrane is comparatively more permeable to K^+ , nearly impermeable to Na^+ , and completely impermeable to negatively charged proteins in axoplasm.
- 2 **Concentration Gradient established:**
 - *Inside Axoplasm:* High K^+ + High negatively charged proteins + Low Na^+ .
 - *Outside ECF:* High Na^+ + Low K^+ .
- 3 **Active Maintenance:** Maintained by active transport via Na^+/K^+ pump (3 Na^+ out / 2 K^+ in). Outer surface positive (+), inner negative (-). Membrane is **Polarised**.

STAGE 2: Depolarisation & Circuit Propagation

- 1 **Stimulus Application (at Site A):** Axonal membrane becomes freely permeable to Na^+ , causing rapid influx of Na^+ .
- 2 **Reversal of Polarity:** Inner side becomes (+), outer becomes (-). Membrane at Site A is **Depolarised**. The electrical potential difference is the **Action Potential (Nerve Impulse)**.
- 3 **Current Circuit Flow:**
 - *Inner surface:* Current flows from **Site A** → **Site B**.
 - *Outer surface:* Current flows from **Site B** → **Site A** (completes circuit).
 - Depolarisation shifts to Site B; sequence repeats along axon length.
- 4 **Repolarisation:** Na^+ permeability is extremely short-lived, followed by immediate rise in K^+ permeability. K^+ diffuses outside, restoring resting potential.

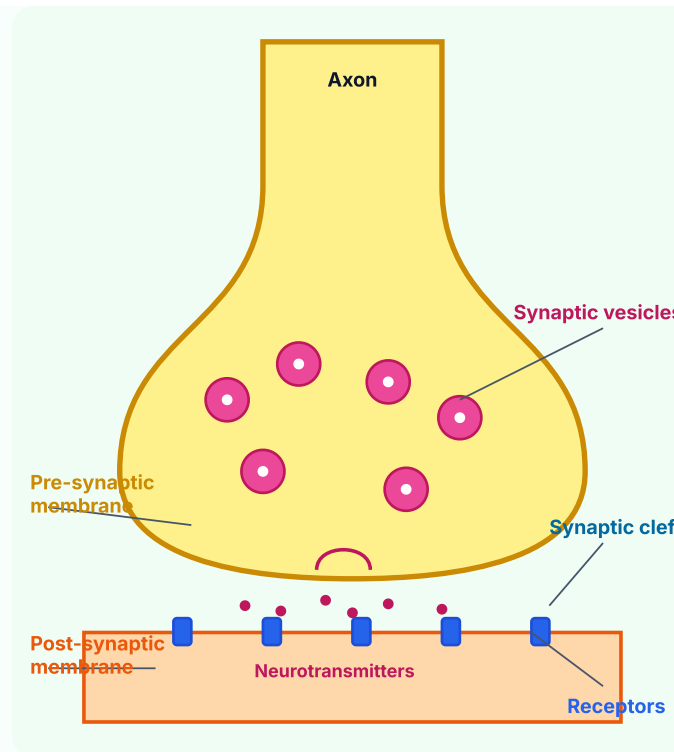
FIGURE 18.2 — DIAGRAMMATIC REPRESENTATION OF IMPULSE CONDUCTION THROUGH AN AXON (SITES A AND B)



18.3.2 TRANSMISSION OF IMPULSES ACROSS SYNAPSE

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FIGURE 18.3 — AXON TERMINAL AND CHEMICAL SYNAPSE



Feature	Electrical Synapse	Chemical Synapse
Synaptic Cleft / Gap	Pre- & post-synaptic membranes in very close proximity (No functional fluid gap).	Separated by a fluid-filled space called Synaptic Cleft .
Mode of Flow	Electrical current flows directly across gap junctions.	Requires chemical messengers called Neurotransmitters .
Transmission Speed	Always faster (resembles conduction along single axon).	Relatively slower due to neurotransmitter release & binding delay.
Abundance in Humans	Rare in the human neural system.	Predominant type in human nervous system.

🔍 SEQUENTIAL MECHANISM OF CHEMICAL SYNAPTIC TRANSMISSION

Step 1

Action potential arrives at the axon terminal.

Step 2

Stimulates movement & fusion of synaptic vesicles with pre-synaptic membrane.

Step 3

Release of neurotransmitters into the synaptic cleft via exocytosis.

Step 4

Neurotransmitters bind to specific **receptors** on post-synaptic membrane.

Step 5

Opens ion channels → Ion entry generates **New Potential** (Excitatory or Inhibitory).

18.4 CENTRAL NEURAL SYSTEM — HUMAN BRAIN ANATOMY & FUNCTIONS

NCERT Pages 235-236

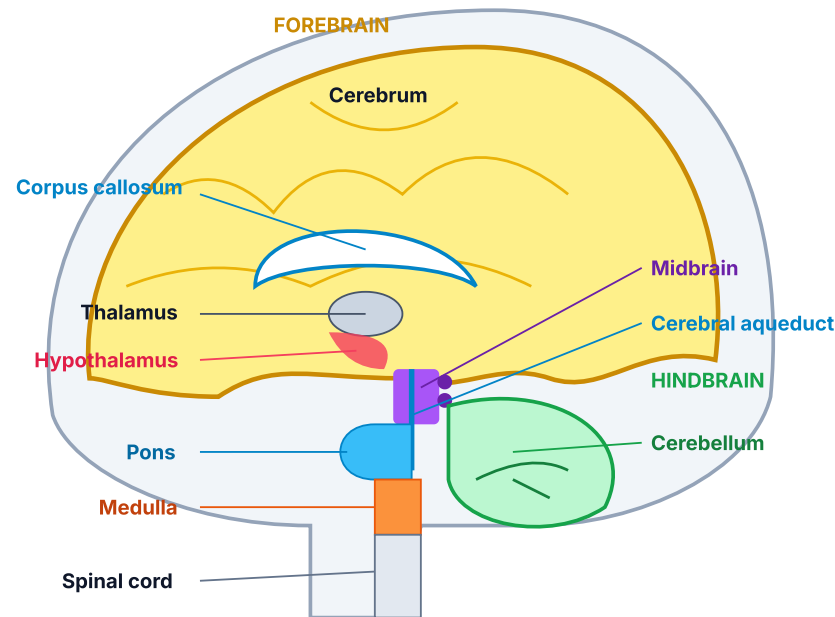
Brain as Command & Control Center

Controls voluntary movements, body balance, functioning of vital involuntary organs (lungs, heart, kidneys), thermoregulation, hunger & thirst, circadian (24-hour) rhythms, endocrine activities, and human behavior. Processing site for vision, hearing, speech, memory, intelligence, emotions, and thoughts.

Cranial Meninges (3 Protective Layers)

1. **Dura mater:** Outer layer (tough & fibrous).
2. **Arachnoid:** Very thin middle layer.
3. **Pia mater:** Inner layer in direct contact with brain tissue.

FIGURE 18.4 — SAGITTAL SECTION OF THE HUMAN BRAIN



1. FOREBRAIN (Cerebrum + Thalamus + Hypothalamus)

- **Cerebrum:** Major part of brain. Longitudinally cleft into Left & Right **Cerebral Hemispheres**, connected by **Corpus Callosum** (tract of nerve fibres).
- **Cerebral Cortex (Grey Matter):** Thrown into prominent folds. Cell bodies concentrated here. Contains:
 - *Motor Areas | Sensory Areas*
 - *Association Areas:* Large regions neither clearly sensory nor motor. Responsible for **intersensory associations, memory, and communication**.
- **White Matter:** Inner cerebral hemisphere made of myelinated nerve fibre tracts.
- **Thalamus:** Major coordinating center for sensory and motor signaling.
- **Hypothalamus:** At base of thalamus. Controls **body temperature, urge for eating & drinking**. Contains neurosecretory cells secreting **hypothalamic hormones**.
- **Limbic System / Limbic Lobe:** Inner cerebral hemispheres + **Amygdala + Hippocampus**. Regulates **sexual behavior, emotional reactions (excitement, pleasure, rage, fear), and motivation**.

2. MIDBRAIN

Located between forebrain (thalamus/hypothalamus) and hindbrain (pons).

- Passageway: **Cerebral Aqueduct** canal passes through it.
- Dorsal portion: 4 round swellings/lobes called **Corpora Quadrigemina**.

3. HINDBRAIN

- **Pons:** Fibre tracts interconnecting different brain regions.
- **Cerebellum:** Very convoluted surface providing extra space for many more neurons.
- **Medulla Oblongata:** Connected to spinal cord. Controls **respiration, cardiovascular reflexes, and gastric secretions.**

🧠 Brain Stem Components:

Midbrain + Pons + Medulla Oblongata

📊 SECTION 7: MASTER ACADEMIC COMPARISON TABLES

100% Bordered Grid Lines

TABLE 18.1 — DENDRITES VS AXONS

Feature	Dendrites	Axons
Number per Neuron	Usually multiple short branching processes.	Usually a single long process.
Nissl's Granules	Present in dendrites.	Absent in axons.
Direction of Impulse	Conducts impulses towards cell body (Centripetal).	Conducts impulses away from cell body (Centrifugal).
Terminal Structure	Fine branching tips without synaptic knobs.	Distal end branches terminate as bulb-like Synaptic Knobs with vesicles.

TABLE 18.2 — MYELINATED VS UNMYELINATED AXONS

Feature	Myelinated Axon	Unmyelinated Axon
Myelin Sheath	Schwann cells form a continuous Myelin Sheath around axon.	Enclosed by Schwann cell that does NOT form myelin sheath.
Nodes of Ranvier	Present (Gaps between adjacent myelin sheaths).	Absent.
Conduction Type	Saltatory conduction (Fast jumping across nodes).	Continuous slow wave conduction.
Anatomical Location	Found in Spinal and Cranial Nerves.	Found in Autonomous and Somatic neural systems.

TABLE 18.3 — RESTING POTENTIAL VS ACTION POTENTIAL

Property	Resting Potential	Action Potential
Membrane State	Polarised State (Unstimulated resting neuron).	Depolarised State (Stimulated neuron).
Charge Distribution	Outer surface = Positive (+) Inner surface = Negative (-) .	Outer surface = Negative (-) Inner surface = Positive (+) .
Ion Permeability	High permeability to K^+ ; Impermeable to Na^+ and proteins.	Freely permeable to Na^+ (Rapid influx).
Potential Value	Approximately -70 mV .	Approximately +30 mV .

TABLE 18.4 — THALAMUS VS HYPOTHALAMUS

Feature	Thalamus	Hypothalamus
Position	Structure wrapped around by cerebrum.	Lies at the base of the thalamus .
Primary Function	Major coordinating centre for sensory and motor signaling .	Controls body temperature, hunger, thirst , and emotional responses.
Endocrine Role	No direct neurosecretory hormone secretion.	Contains neurosecretory cells secreting Hypothalamic Hormones .

TABLE 18.5 — CEREBRUM VS CEREBELLUM

Feature	Cerebrum	Cerebellum
Brain Division	Part of the Forebrain (Major portion of brain).	Part of the Hindbrain .
Surface Features	Has prominent folds forming gyri and sulci .	Has a very convoluted surface for extra neuron space.
Main Function	Memory, intelligence, voluntary movement, association.	Involuntary muscular coordination & body balance maintenance.

TABLE 18.6 — CENTRAL NEURAL SYSTEM (CNS) VS PERIPHERAL NEURAL SYSTEM (PNS)

Feature	CNS	PNS
Components	Brain + Spinal Cord.	All cranial and spinal nerves associated with CNS.
Primary Function	Site of information processing and control .	Transmits sensory/motor impulses between body and CNS.

SECTION 8: MEMORY TRICKS & HIGH-YIELD EXAM HOTSPOTS

NEET / Board Quick Recall

Trick 1: Cranial Meninges (Outer to Inner)

Mnemonic: "D-A-P" (Pad from inside, DAP from outside)

- **D** = **Dura mater** (Outer layer)
- **A** = **Arachnoid** (Middle thin layer)
- **P** = **Pia mater** (Inner layer touching tissue)

Trick 2: Neuron Locations (Unipolar, Bipolar, Multipolar)

Mnemonic: "E-R-C" (Embryo, Retina, Cortex)

- **Unipolar** = Embryonic stage
- **Bipolar** = Retina of eye
- **Multipolar** = Cerebral Cortex

Trick 3: Brain Stem Components

Mnemonic: "M-P-M" (Midbrain, Pons, Medulla)

Remember: Cerebellum is *NOT* part of the brain stem! Only Midbrain + Pons + Medulla oblongata form the Brain Stem.

High-Yield NEET Exam Hotspots:

- **Nissl's granules location:** Present in Cell Body (Soma) and Dendrites, but strictly **ABSENT in Axons**.
- **Sodium-Potassium Pump:** Transports **3 Na^+ outwards for 2 K^+ inwards** using ATP. This pump maintains the negative resting potential.
- **Association Areas:** Located in cerebral cortex; neither sensory nor motor in function; responsible for **intersensory associations, memory, and communication**.

- **Hypothalamus Centers:** Controls thermoregulation (body temperature), hunger, thirst, and neurosecretory hormone release.
- **Limbic System:** Amygdala + Hippocampus + Inner Cerebrum + Hypothalamus → controls sexual behavior, emotional reactions (rage, fear, pleasure), and motivation.

SECTION 9: COMPLETE NCERT CHAPTER EXERCISES & OFFICIAL SOLUTIONS

All 10 Questions Solved

Q1. Briefly describe the structure of the Brain.

Answer: The human brain is protected by the skull and covered by three cranial meninges: outer *dura mater*, middle *arachnoid*, and inner *pia mater*. It is divided into three major parts:

1. **Forebrain:** Consists of *Cerebrum* (left & right hemispheres connected by corpus callosum; cortex contains motor, sensory & association areas), *Thalamus* (relay center), and *Hypothalamus* (thermoregulation, hunger, thirst, hormone secretion). Limbic system controls emotions/motivation.
2. **Midbrain:** Located between forebrain and hindbrain; contains cerebral aqueduct and *corpora quadrigemina* (4 optic/auditory lobes).
3. **Hindbrain:** Consists of *Pons* (fibre tracts), *Cerebellum* (convoluted, controls muscular balance), and *Medulla Oblongata* (controls respiration, cardiovascular reflexes, gastric secretions).

Q2. Compare the following: (a) Central neural system (CNS) and Peripheral neural system (PNS) (b) Resting potential and action potential

Answer:

(a) *CNS vs PNS:* CNS comprises brain and spinal cord and acts as the central site for information processing and control. PNS comprises all nerves associated with CNS, categorized into afferent/efferent and somatic/autonomic divisions.

(b) *Resting vs Action Potential:* Resting potential is the electrical potential (-70mV) across an unstimulated polarised axonal membrane (outer +, inner -). Action potential (+30mV) is the temporary reversal of polarity (outer -, inner +) caused by rapid Na^+ influx upon stimulus application.

Q3. Explain the following processes: (a) Polarisation of the membrane of a nerve fibre (b) Depolarisation of the membrane of a nerve fibre (c) Transmission of a nerve impulse across a chemical synapse

Answer:

(a) **Polarisation:** In resting state, axonal membrane is more permeable to K^+ and impermeable to Na^+ and negative proteins.

Na^+/K^+ pump transports 3Na^+ out for 2K^+ in, creating a positive outer surface and negative inner surface (-70mV).

(b) **Depolarisation:** Stimulus causes sudden increase in Na^+ permeability. Influx of Na^+ reverses polarity at that site (inner +, outer -), generating an action potential.

(c) **Synaptic Transmission:** Action potential arriving at axon terminal causes synaptic vesicles to fuse with pre-synaptic membrane, releasing neurotransmitters into the synaptic cleft. They bind to post-synaptic receptors, opening ion channels to generate a new potential.

Q4. Draw labelled diagrams of the following: (a) Neuron (b) Brain

Answer: Please refer to **Figure 18.1** (Neuron Diagram in Section 18.3) and **Figure 18.4** (Sagittal Section of Brain Diagram in Section 18.4) above for complete high-resolution labelled vector illustrations.

Q5. Write short notes on the following: (a) Neural coordination (b) Forebrain (c) Midbrain (d) Hindbrain (e) Synapse

Answer:

- (a) *Neural coordination*: Rapid process of interaction between organs to complement functions and maintain homeostasis via organized point-to-point connections.
- (b) *Forebrain*: Consists of cerebrum, thalamus, and hypothalamus; controls higher cognitive functions, emotions, body temp, and hunger.
- (c) *Midbrain*: Integrates visual, tactile, and auditory inputs; contains corpora quadrigemina and cerebral aqueduct.
- (d) *Hindbrain*: Comprises pons, cerebellum, and medulla; regulates balance, cardiovascular reflexes, gastric secretions, and respiration.
- (e) *Synapse*: Functional junction between pre- and post-synaptic neurons (electrical or chemical via synaptic cleft).

Q6. Give a brief account of Mechanism of synaptic transmission.

Answer: Arriving action potential → Synaptic vesicle translocation → Fusion with pre-synaptic membrane → Exocytosis of neurotransmitters into synaptic cleft → Binding to post-synaptic receptors → Ion channel opening → Generation of excitatory/inhibitory potential.

Q7. Explain the role of Na^+ in the generation of action potential.

Answer: Upon stimulation, the axonal membrane becomes freely permeable to Na^+ . The rapid influx of Na^+ down its concentration gradient neutralizes negative internal charges and turns the axoplasm positive relative to the outside, generating the action potential.

Q8. Differentiate between: (a) Myelinated and non-myelinated axons (b) Dendrites and axons (c) Thalamus and Hypothalamus (d) Cerebrum and Cerebellum

Answer: Please refer to **Section 7 (Master Comparison Tables 18.1, 18.2, 18.4, 18.5)** for complete tabular comparisons.

Q9. Answer the following: (a) Which part of the human brain is the most developed? (b) Which part of our central neural system acts as a master clock?

Answer:

- (a) The **Cerebrum** (specifically the Cerebral Cortex) is the most developed part of the human brain.
- (b) The **Hypothalamus** (specifically the Suprachiasmatic nucleus) acts as the master clock controlling 24-hour circadian rhythms.

Q10. Distinguish between: (a) afferent neurons and efferent neurons (b) impulse conduction in a myelinated nerve fibre and unmyelinated nerve fibre (c) cranial nerves and spinal nerves

Answer:

- (a) *Afferent vs Efferent*: Afferent carries sensory impulses from organs to CNS; Efferent carries motor regulatory commands from CNS to effectors.
- (b) *Myelinated vs Unmyelinated Conduction*: Myelinated uses saltatory conduction (fast jumping between Nodes of Ranvier); Unmyelinated uses continuous wave conduction (slower).
- (c) *Cranial vs Spinal Nerves*: Cranial nerves originate directly from the brain (12 pairs in humans); Spinal nerves originate from the spinal cord (31 pairs).



FINAL RAPID REVISION BOX

1-Minute Pre-Exam Scan

Polarization & Resting State

Axonal membrane permeable to K^+ , impermeable to Na^+ . Na^+/K^+ pump active (3 Na^+ out / 2 K^+ in).
Potential = -70mV.

Action Potential

Stimulus at Site A $\rightarrow$ Na^+ influx $\rightarrow$ Depolarization (+30mV). Current flows inner A $\rightarrow$ B, outer B $\rightarrow$ A.

Synapses

Electrical = faster, rare. Chemical = uses neurotransmitters in synaptic vesicles binding to post-synaptic receptors.

Brain Divisions

Forebrain (Cerebrum, Thalamus, Hypothalamus), Midbrain (Corpora Quadrigemina), Hindbrain (Pons, Cerebellum, Medulla).